

Factors and Multiples

Divisibility, prime numbers, factor pairs—the building blocks of number theory.



Use these notes well and you will be able to:

- Define factors and multiples and explain the difference.
- List all factors of any whole number systematically.
- Identify prime and composite numbers.
- Use prime factor decomposition (factor tree and division methods).
- Apply tests of divisibility for 2, 3, 4, 5, 6, 8, 9, 10.

If you follow the steps, you will succeed.

Key Words

KEY WORD	SIMPLE DEFINITION
Factor	A whole number that divides exactly into another with no remainder.
Multiple	The result of multiplying a number by a positive integer.
Factor pair	Two numbers that multiply together to give a target. e.g. $3 \times 4 = 12$.
Prime number	Has exactly TWO factors: 1 and itself. e.g. 2, 3, 5, 7, 11.
Composite number	Has MORE than two factors — not prime. e.g. 4, 6, 9, 12.
1	NEITHER prime nor composite — it has only one factor (itself).
Prime factorisation	Writing a number as a product of its prime factors only.
Factor tree	A diagram that breaks a number into factor pairs until all are prime.
Divisibility rule	A quick test to decide if a number is a factor without dividing.
Common factor	A number that is a factor of TWO or more numbers.
Common multiple	A number that is a multiple of TWO or more numbers.

Factors — Systematic Listing Method

To find ALL factors of a number: start at 1, test each integer in order. Write factor PAIRS — when the two numbers in a pair are the same or cross over, stop. This ensures no factor is missed.

List all factors of 36

1×36 factors: 1, 36

2×18 factors: 2, 18

3×12 factors: 3, 12

4×9 factors: 4, 9

$5 \times ?$ 5 does not divide 36 exactly — skip

6×6 factor: 6 (same number — STOP)

Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36 (9 factors total)

Multiples — Listing and Identifying

Multiples of n are the results of $n \times 1, n \times 2, n \times 3, \dots$ They continue infinitely.

NUMBER	FIRST TEN MULTIPLES
3	3, 6, 9, 12, 15, 18, 21, 24, 27, 30
4	4, 8, 12, 16, 20, 24, 28, 32, 36, 40
7	7, 14, 21, 28, 35, 42, 49, 56, 63, 70
12	12, 24, 36, 48, 60, 72, 84, 96, 108, 120

Divisibility Rules

÷ BY	RULE	EXAMPLE
2	Last digit is even (0,2,4,6,8)	346 6 is even
3	Sum of digits divisible by 3	252 $2+5+2=9, 9 \div 3=3$
4	Last TWO digits divisible by 4	316 $16 \div 4=4$
5	Ends in 0 or 5	235 ends in 5
6	Divisible by BOTH 2 and 3	312 even, $3+1+2=6$
8	Last THREE digits divisible by 8	1048 $048 \div 8=6$
9	Sum of digits divisible by 9	729 $7+2+9=18, 18 \div 9=2$
10	Ends in 0	450 ends in 0

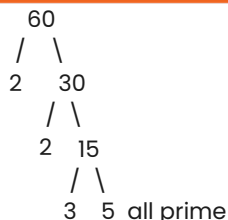
Prime Numbers

A prime number has EXACTLY two factors: 1 and itself. 1 is NOT prime (only one factor). 2 is the only EVEN prime.

Primes up to 100: 2,3,5,7,11,13,17,19,23,29,31,37,41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

Prime Factor Decomposition — Two Methods

Method A: Factor Tree (for 60)



Result: $60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

Method B: Repeated Division (ladder method) for 60

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2 | 60
2 | 30
3 | 15
5 | 5
  1 stop when you reach 1
  
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Read primes down the left: $60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

Write 180 as a product of prime factors

$180 \div 2 = 90$ (2 is prime)
 $90 \div 2 = 45$ (2 is prime)
 $45 \div 3 = 15$ (3 is prime)
 $15 \div 3 = 5$ (3 is prime)
 $5 \div 5 = 1$ (5 is prime — stop)
 $180 = 2^2 \times 3^2 \times 5$

Practice Questions

FACTORS AND MULTIPLES

- List all factors of 48.
- List the first eight multiples of 9.
- Is 315 divisible by 3? By 5? By 9? Use divisibility rules — show digit sums.
- Which numbers below 30 are prime? List them.

PRIME FACTORISATION

- Write 72 as a product of prime factors using a factor tree.
- Write 360 as a product of prime factors using repeated division.
- A number written in prime factor form is $2^3 \times 3 \times 7$. What is the number?
- Show that 84 and 126 have 2×3 as a common prime factor.

ADD YOUR NOTES HERE**Hey smart student!**

Factors and primes are the atoms of number theory. Everything in number work — HCF, LCM, fractions — comes back to prime factors.

Memorise primes up to 50 and prime factorisations become fast.

Keep going — see you in the next lesson! From FREDi